

Arithmetic Sequence Word Problems

Directions: For each situation, write an explicit formula for the arithmetic sequence and then answer the question. Show all work.

1. A warehouse packs 18 boxes during the first hour and 5 more boxes during each later hour. How many boxes are packed during hour 14?
2. A tile design has 7 tiles in its first row. Each new row has 4 more tiles than the previous row. How many tiles are in row 20?
3. A runner completes 12 miles during the first week of a training plan and adds 3 miles each week. How many miles does the runner complete during week 9?
4. A freezer temperature is 18°C at the first hourly reading and decreases by 4°C each hour. What is the temperature at the eighth reading?

5. A display has 50 cans in its first row. Each row above it has 3 fewer cans than the row below it. How many cans are in row 12?
6. A plant is 15 cm tall at the end of week 1 and grows 2 cm each week. How tall is the plant at the end of week 16?
7. A parking garage has 140 spaces on level 1. Each higher level has 12 fewer spaces than the level below it. How many spaces are on level 8?
8. A student completes 24 practice problems on day 1 and increases the number completed by 6 each day. How many problems are completed on day 11?

9. A marching band places 10 students in the first row and 2 more students in each row after that. How many students are in row 17?

10. A water tank contains 260 gallons at the first measurement. The amount decreases by 15 gallons at each later measurement. How many gallons are in the 13th measurement?